

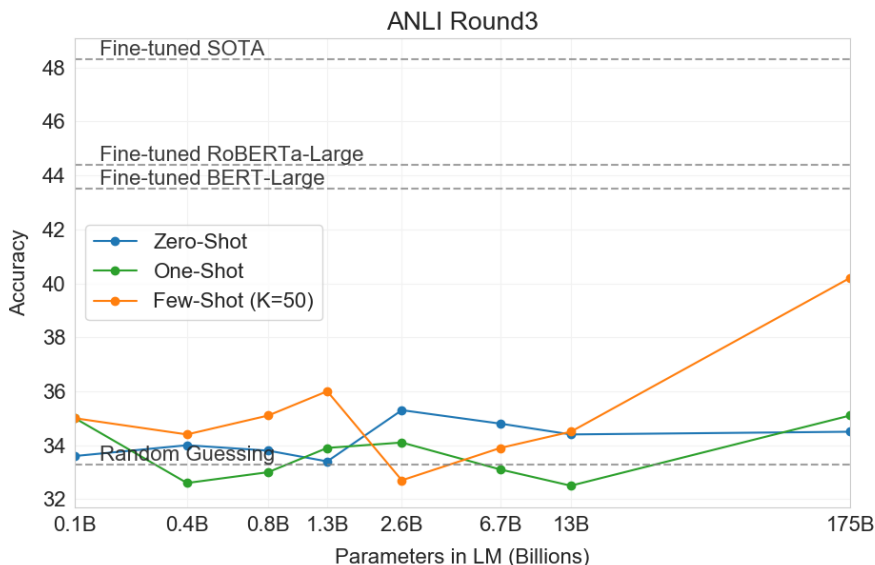


Figure 3.9: Performance of GPT-3 on ANLI Round 3. Results are on the dev-set, which has only 1500 examples and therefore has high variance (we estimate a standard deviation of 1.2%). We find that smaller models hover around random chance, while few-shot GPT-3 175B closes almost half the gap from random chance to SOTA. Results for ANLI rounds 1 and 2 are shown in the appendix.

whether the second sentence logically follows from the first, contradicts the first sentence, or is possibly true (neutral). SuperGLUE includes an NLI dataset, RTE, which evaluates the binary version of the task. On RTE, only the largest version of GPT-3 performs convincingly better than random (56%) in any evaluation setting, but in a few-shot setting GPT-3 performs similarly to a single-task fine-tuned BERT Large. We also evaluate on the recently introduced Adversarial Natural Language Inference (ANLI) dataset [NWD⁺19]. ANLI is a difficult dataset employing a series of adversarially mined natural language inference questions in three rounds (R1, R2, and R3). Similar to RTE, all of our models smaller than GPT-3 perform at almost exactly random chance on ANLI, even in the few-shot setting ($\sim 33\%$), whereas GPT-3 itself shows signs of life on Round 3. Results for ANLI R3 are highlighted in Figure 3.9 and full results for all rounds can be found in Appendix H. These results on both RTE and ANLI suggest that NLI is still a very difficult task for language models and they are only just beginning to show signs of progress.

3.9 Synthetic and Qualitative Tasks

One way to probe GPT-3’s range of abilities in the few-shot (or zero- and one-shot) setting is to give it tasks which require it to perform simple on-the-fly computational reasoning, recognize a novel pattern that is unlikely to have occurred in training, or adapt quickly to an unusual task. We devise several tasks to test this class of abilities. First, we test GPT-3’s ability to perform arithmetic. Second, we create several tasks that involve rearranging or unscrambling the letters in a word, tasks which are unlikely to have been exactly seen during training. Third, we test GPT-3’s ability to solve SAT-style analogy problems few-shot. Finally, we test GPT-3 on several qualitative tasks, including using new words in a sentence, correcting English grammar, and news article generation. We will release the synthetic datasets with the hope of stimulating further study of test-time behavior of language models.

3.9.1 Arithmetic

To test GPT-3’s ability to perform simple arithmetic operations without task-specific training, we developed a small battery of 10 tests that involve asking GPT-3 a simple arithmetic problem in natural language:

- **2 digit addition (2D+)** – The model is asked to add two integers sampled uniformly from $[0, 100)$, phrased in the form of a question, e.g. “Q: What is 48 plus 76? A: 124.”
- **2 digit subtraction (2D-)** – The model is asked to subtract two integers sampled uniformly from $[0, 100)$; the answer may be negative. Example: “Q: What is 34 minus 53? A: -19”.
- **3 digit addition (3D+)** – Same as 2 digit addition, except numbers are uniformly sampled from $[0, 1000)$.

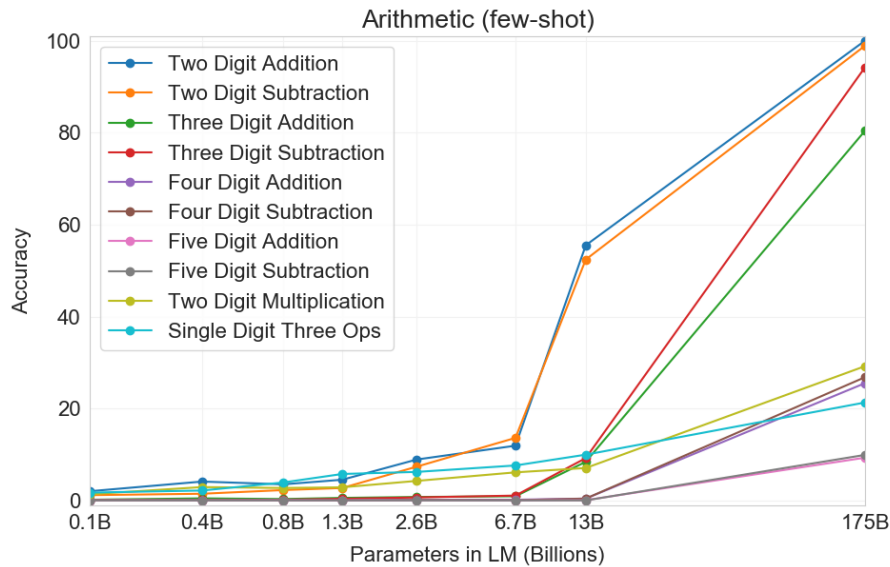


Figure 3.10: Results on all 10 arithmetic tasks in the few-shot settings for models of different sizes. There is a significant jump from the second largest model (GPT-3 13B) to the largest model (GPT-3 175), with the latter being able to reliably accurate 2 digit arithmetic, usually accurate 3 digit arithmetic, and correct answers a significant fraction of the time on 4-5 digit arithmetic, 2 digit multiplication, and compound operations. Results for one-shot and zero-shot are shown in the appendix.

- **3 digit subtraction (3D-)** – Same as 2 digit subtraction, except numbers are uniformly sampled from $[0, 1000)$.
- **4 digit addition (4D+)** – Same as 3 digit addition, except uniformly sampled from $[0, 10000)$.
- **4 digit subtraction (4D-)** – Same as 3 digit subtraction, except uniformly sampled from $[0, 10000)$.
- **5 digit addition (5D+)** – Same as 3 digit addition, except uniformly sampled from $[0, 100000)$.
- **5 digit subtraction (5D-)** – Same as 3 digit subtraction, except uniformly sampled from $[0, 100000)$.
- **2 digit multiplication (2Dx)** – The model is asked to multiply two integers sampled uniformly from $[0, 100)$, e.g. “Q: What is 24 times 42? A: 1008”.
- **One-digit composite (1DC)** – The model is asked to perform a composite operation on three 1 digit numbers, with parentheses around the last two. For example, “Q: What is $6+(4*8)$? A: 38”. The three 1 digit numbers are selected uniformly on $[0, 10)$ and the operations are selected uniformly from $\{+, -, *\}$.

In all 10 tasks the model must generate the correct answer exactly. For each task we generate a dataset of 2,000 random instances of the task and evaluate all models on those instances.

First we evaluate GPT-3 in the few-shot setting, for which results are shown in Figure 3.10. On addition and subtraction, GPT-3 displays strong proficiency when the number of digits is small, achieving 100% accuracy on 2 digit addition, 98.9% at 2 digit subtraction, 80.2% at 3 digit addition, and 94.2% at 3-digit subtraction. Performance decreases as the number of digits increases, but GPT-3 still achieves 25-26% accuracy on four digit operations and 9-10% accuracy on five digit operations, suggesting at least some capacity to generalize to larger numbers of digits. GPT-3 also achieves 29.2% accuracy at 2 digit multiplication, an especially computationally intensive operation. Finally, GPT-3 achieves 21.3% accuracy at single digit combined operations (for example, $9*(7+5)$), suggesting that it has some robustness beyond just single operations.

As Figure 3.10 makes clear, small models do poorly on all of these tasks – even the 13 billion parameter model (the second largest after the 175 billion full GPT-3) can solve 2 digit addition and subtraction only half the time, and all other operations less than 10% of the time.

One-shot and zero-shot performance are somewhat degraded relative to few-shot performance, suggesting that adaptation to the task (or at the very least recognition of the task) is important to performing these computations correctly. Nevertheless, one-shot performance is still quite strong, and even zero-shot performance of the full GPT-3 significantly

Setting	2D+	2D-	3D+	3D-	4D+	4D-	5D+	5D-	2Dx	1DC
GPT-3 Zero-shot	76.9	58.0	34.2	48.3	4.0	7.5	0.7	0.8	19.8	9.8
GPT-3 One-shot	99.6	86.4	65.5	78.7	14.0	14.0	3.5	3.8	27.4	14.3
GPT-3 Few-shot	100.0	98.9	80.4	94.2	25.5	26.8	9.3	9.9	29.2	21.3

Table 3.9: Results on basic arithmetic tasks for GPT-3 175B. $\{2,3,4,5\}D\{+,-\}$ is 2, 3, 4, and 5 digit addition or subtraction, 2Dx is 2 digit multiplication. 1DC is 1 digit composite operations. Results become progressively stronger moving from the zero-shot to one-shot to few-shot setting, but even the zero-shot shows significant arithmetic abilities.

Setting	CL	A1	A2	RI	RW
GPT-3 Zero-shot	3.66	2.28	8.91	8.26	0.09
GPT-3 One-shot	21.7	8.62	25.9	45.4	0.48
GPT-3 Few-shot	37.9	15.1	39.7	67.2	0.44

Table 3.10: GPT-3 175B performance on various word unscrambling and word manipulation tasks, in zero-, one-, and few-shot settings. CL is “cycle letters in word”, A1 is anagrams of but the first and last letters, A2 is anagrams of all but the first and last two letters, RI is “Random insertion in word”, RW is “reversed words”.

outperforms few-shot learning for all smaller models. All three settings for the full GPT-3 are shown in Table 3.9, and model capacity scaling for all three settings is shown in Appendix H.

To spot-check whether the model is simply memorizing specific arithmetic problems, we took the 3-digit arithmetic problems in our test set and searched for them in our training data in both the forms “<NUM1> + <NUM2> =” and “<NUM1> plus <NUM2>”. Out of 2,000 addition problems we found only 17 matches (0.8%) and out of 2,000 subtraction problems we found only 2 matches (0.1%), suggesting that only a trivial fraction of the correct answers could have been memorized. In addition, inspection of incorrect answers reveals that the model often makes mistakes such as not carrying a “1”, suggesting it is actually attempting to perform the relevant computation rather than memorizing a table.

Overall, GPT-3 displays reasonable proficiency at moderately complex arithmetic in few-shot, one-shot, and even zero-shot settings.

3.9.2 Word Scrambling and Manipulation Tasks

To test GPT-3’s ability to learn novel symbolic manipulations from a few examples, we designed a small battery of 5 “character manipulation” tasks. Each task involves giving the model a word distorted by some combination of scrambling, addition, or deletion of characters, and asking it to recover the original word. The 5 tasks are:

- **Cycle letters in word (CL)** – The model is given a word with its letters cycled, then the “=” symbol, and is expected to generate the original word. For example, it might be given “lyinevitab” and should output “inevitably”.
- **Anagrams of all but first and last characters (A1)** – The model is given a word where every letter except the first and last have been scrambled randomly, and must output the original word. Example: criroptuon = corruption.
- **Anagrams of all but first and last 2 characters (A2)** – The model is given a word where every letter except the first 2 and last 2 have been scrambled randomly, and must recover the original word. Example: opoepnnt → opponent.
- **Random insertion in word (RI)** – A random punctuation or space character is inserted between each letter of a word, and the model must output the original word. Example: s.u!c/c!e.s s i/o/n = succession.
- **Reversed words (RW)** – The model is given a word spelled backwards, and must output the original word. Example: stcejbo → objects.

For each task we generate 10,000 examples, which we chose to be the top 10,000 most frequent words as measured by [Nor09] of length more than 4 characters and less than 15 characters. The few-shot results are shown in Figure 3.11. Task performance tends to grow smoothly with model size, with the full GPT-3 model achieving 66.9% on removing

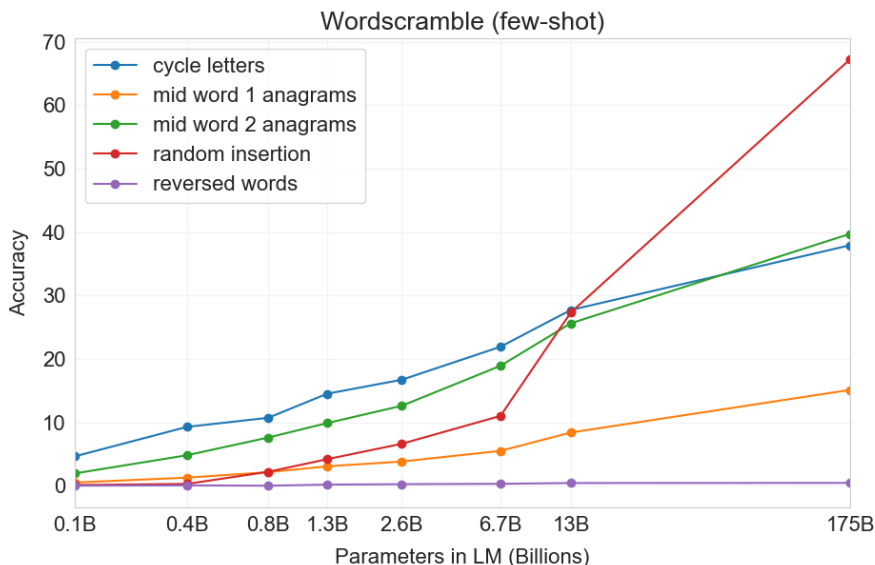


Figure 3.11: Few-shot performance on the five word scrambling tasks for different sizes of model. There is generally smooth improvement with model size although the random insertion task shows an upward slope of improvement with the 175B model solving the majority of the time. Scaling of one-shot and zero-shot performance is shown in the appendix. All tasks are done with $K = 100$.

random insertions, 38.6% on cycling letters, 40.2% on the easier anagram task, and 15.1% on the more difficult anagram task (where only the first and last letters are held fixed). None of the models can reverse the letters in a word.

In the one-shot setting, performance is significantly weaker (dropping by half or more), and in the zero-shot setting the model can rarely perform any of the tasks (Table 3.10). This suggests that the model really does appear to learn these tasks at test time, as the model cannot perform them zero-shot and their artificial nature makes them unlikely to appear in the pre-training data (although we cannot confirm this with certainty).

We can further quantify performance by plotting “in-context learning curves”, which show task performance as a function of the number of in-context examples. We show in-context learning curves for the Symbol Insertion task in Figure 1.2. We can see that larger models are able to make increasingly effective use of in-context information, including both task examples and natural language task descriptions.

Finally, it is worth adding that solving these tasks requires character-level manipulations, whereas our BPE encoding operates on significant fractions of a word (on average ~ 0.7 words per token), so from the LM’s perspective succeeding at these tasks involves not just manipulating BPE tokens but understanding and pulling apart their substructure. Also, CL, A1, and A2 are not bijective (that is, the unscrambled word is not a deterministic function of the scrambled word), requiring the model to perform some search to find the correct unscrambling. Thus, the skills involved appear to require non-trivial pattern-matching and computation.

3.9.3 SAT Analogies

To test GPT-3 on another task that is somewhat unusual relative to the typical distribution of text, we collected a set of 374 “SAT analogy” problems [TLBS03]. Analogies are a style of multiple choice question that constituted a section of the SAT college entrance exam before 2005. A typical example is “audacious is to boldness as (a) sanctimonious is to hypocrisy, (b) anonymous is to identity, (c) remorseful is to misdeed, (d) deleterious is to result, (e) impressionable is to temptation”. The student is expected to choose which of the five word pairs has the same relationship as the original word pair; in this example the answer is “sanctimonious is to hypocrisy”. On this task GPT-3 achieves 65.2% in the few-shot setting, 59.1% in the one-shot setting, and 53.7% in the zero-shot setting, whereas the average score among college applicants was 57% [TL05] (random guessing yields 20%). As shown in Figure 3.12, the results improve with scale, with the the full 175 billion model improving by over 10% compared to the 13 billion parameter model.

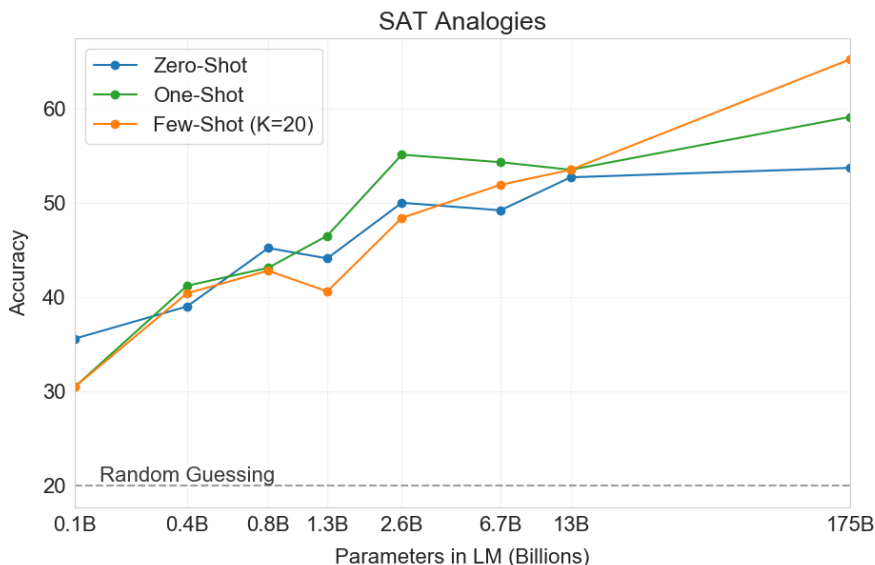


Figure 3.12: Zero-, one-, and few-shot performance on SAT analogy tasks, for different sizes of model. The largest model achieves 65% accuracy in the few-shot setting, and also demonstrates significant gains to in-context learning which are not present in smaller models.

3.9.4 News Article Generation

Previous work on generative language models qualitatively tested their ability to generate synthetic “news articles” by conditional sampling from the model given a human-written prompt consisting of a plausible first sentence for a news story [RWC⁺19]. Relative to [RWC⁺19], the dataset used to train GPT-3 is much less weighted towards news articles, so trying to generate news articles via raw unconditional samples is less effective – for example GPT-3 often interprets the proposed first sentence of a “news article” as a tweet and then posts synthetic responses or follow-up tweets. To solve this problem we employed GPT-3’s few-shot learning abilities by providing three previous news articles in the model’s context to condition it. With the title and subtitle of a proposed next article, the model is able to reliably generate short articles in the “news” genre.

To gauge the quality of news article generation from GPT-3 (which we believe is likely to be correlated with conditional sample generation quality in general), we decided to measure human ability to distinguish GPT-3-generated articles from real ones. Similar work has been carried out by Kreps et al. [KMB20] and Zellers et al. [ZHR⁺19]. Generative language models are trained to match the distribution of content generated by humans, so the (in)ability of humans to distinguish the two is a potentially important measure of quality.³

In order to see how well humans can detect model generated text, we arbitrarily selected 25 article titles and subtitles from the website [newser.com](https://www.newser.com) (mean length: 215 words). We then generated completions of these titles and subtitles from four language models ranging in size from 125M to 175B (GPT-3) parameters (mean length: 200 words). For each model, we presented around 80 US-based participants with a quiz consisting of these real titles and subtitles followed by either the human written article or the article generated by the model⁴. Participants were asked to select whether the article was “very likely written by a human”, “more likely written by a human”, “I don’t know”, “more likely written by a machine”, or “very likely written by a machine”.

The articles we selected were not in the models’ training data and the model outputs were formatted and selected programmatically to prevent human cherry-picking. All models used the same context to condition outputs on and were pre-trained with the same context size and the same article titles and subtitles were used as prompts for each model. However, we also ran an experiment to control for participant effort and attention that followed the same format but involved intentionally bad model generated articles. This was done by generating articles from a “control model”: a 160M parameter model with no context and increased output randomness.

³This task is also relevant to the potential misuse of language models discussed in Section 6.1.

⁴We wanted to identify how good an average person on the internet is at detecting language model outputs, so we focused on participants drawn from the general US population. See Appendix E for details.

	Mean accuracy	95% Confidence Interval (low, hi)	t compared to control (p -value)	“I don’t know” assignments
Control (deliberately bad model)	86%	83%–90%	-	3.6 %
GPT-3 Small	76%	72%–80%	3.9 ($2e-4$)	4.9%
GPT-3 Medium	61%	58%–65%	10.3 ($7e-21$)	6.0%
GPT-3 Large	68%	64%–72%	7.3 ($3e-11$)	8.7%
GPT-3 XL	62%	59%–65%	10.7 ($1e-19$)	7.5%
GPT-3 2.7B	62%	58%–65%	10.4 ($5e-19$)	7.1%
GPT-3 6.7B	60%	56%–63%	11.2 ($3e-21$)	6.2%
GPT-3 13B	55%	52%–58%	15.3 ($1e-32$)	7.1%
GPT-3 175B	52%	49%–54%	16.9 ($1e-34$)	7.8%

Table 3.11: Human accuracy in identifying whether short (~ 200 word) news articles are model generated. We find that human accuracy (measured by the ratio of correct assignments to non-neutral assignments) ranges from 86% on the control model to 52% on GPT-3 175B. This table compares mean accuracy between five different models, and shows the results of a two-sample T-Test for the difference in mean accuracy between each model and the control model (an unconditional GPT-3 Small model with increased output randomness).

Mean human accuracy (the ratio of correct assignments to non-neutral assignments per participant) at detecting that the intentionally bad articles were model generated was $\sim 86\%$ where 50% is chance level performance. By contrast, mean human accuracy at detecting articles that were produced by the 175B parameter model was barely above chance at $\sim 52\%$ (see Table 3.11).⁵ Human abilities to detect model generated text appear to decrease as model size increases: there appears to be a trend towards chance accuracy with model size, and human detection of GPT-3 is close to chance.⁶ This is true despite the fact that participants spend more time on each output as model size increases (see Appendix E).

Examples of synthetic articles from GPT-3 are given in Figures 3.14 and 3.15.⁷ Much of the text is—as indicated by the evaluations—difficult for humans to distinguish from authentic human content. Factual inaccuracies can be an indicator that an article is model generated since, unlike human authors, the models have no access to the specific facts that the article titles refer to or when the article was written. Other indicators include repetition, non sequiturs, and unusual phrasings, though these are often subtle enough that they are not noticed.

Related work on language model detection by Ippolito et al. [IDCBE19] indicates that automatic discriminators like GROVER [ZHR⁺19] and GLTR [GSR19] may have greater success at detecting model generated text than human evaluators. Automatic detection of these models may be a promising area of future research.

Ippolito et al. [IDCBE19] also note that human accuracy at detecting model generated text increases as humans observe more tokens. To do a preliminary investigation of how good humans are at detecting longer news articles generated by GPT-3 175B, we selected 12 world news articles from Reuters with an average length of 569 words and generated completions of these articles from GPT-3 with an average length of 498 words (298 words longer than our initial experiments). Following the methodology above, we ran two experiments, each on around 80 US-based participants, to compare human abilities to detect the articles generated by GPT-3 and a control model.

We found that mean human accuracy at detecting the intentionally bad longer articles from the control model was $\sim 88\%$, while mean human accuracy at detecting the longer articles that were produced by GPT-3 175B was still barely above chance at $\sim 52\%$ (see Table 3.12). This indicates that, for news articles that are around 500 words long, GPT-3 continues to produce articles that humans find difficult to distinguish from human written news articles.

3.9.5 Learning and Using Novel Words

A task studied in developmental linguistics [CB78] is the ability to learn and utilize new words, for example using a word in a sentence after seeing it defined only once, or conversely inferring a word’s meaning from only one usage. Here we qualitatively test GPT-3’s ability to do the former. Specifically, we give GPT-3 the definition of a nonexistent word, such as “Gigamuru”, and then ask it to use it in a sentence. We provide one to five previous examples of a (separate)

⁵We use a two-sample Student’s T-Test to test for significant difference between the means of the participant accuracies of each model and the control model and report the normalized difference in the means (as the t -statistic) and the p -value.

⁶If a model consistently produces texts that are more impressive than human articles, it is possible that human performance on this task would drop below 50%. Indeed, many individual participants scored below 50% on this task.

⁷Additional non-news samples can be found in Appendix F.

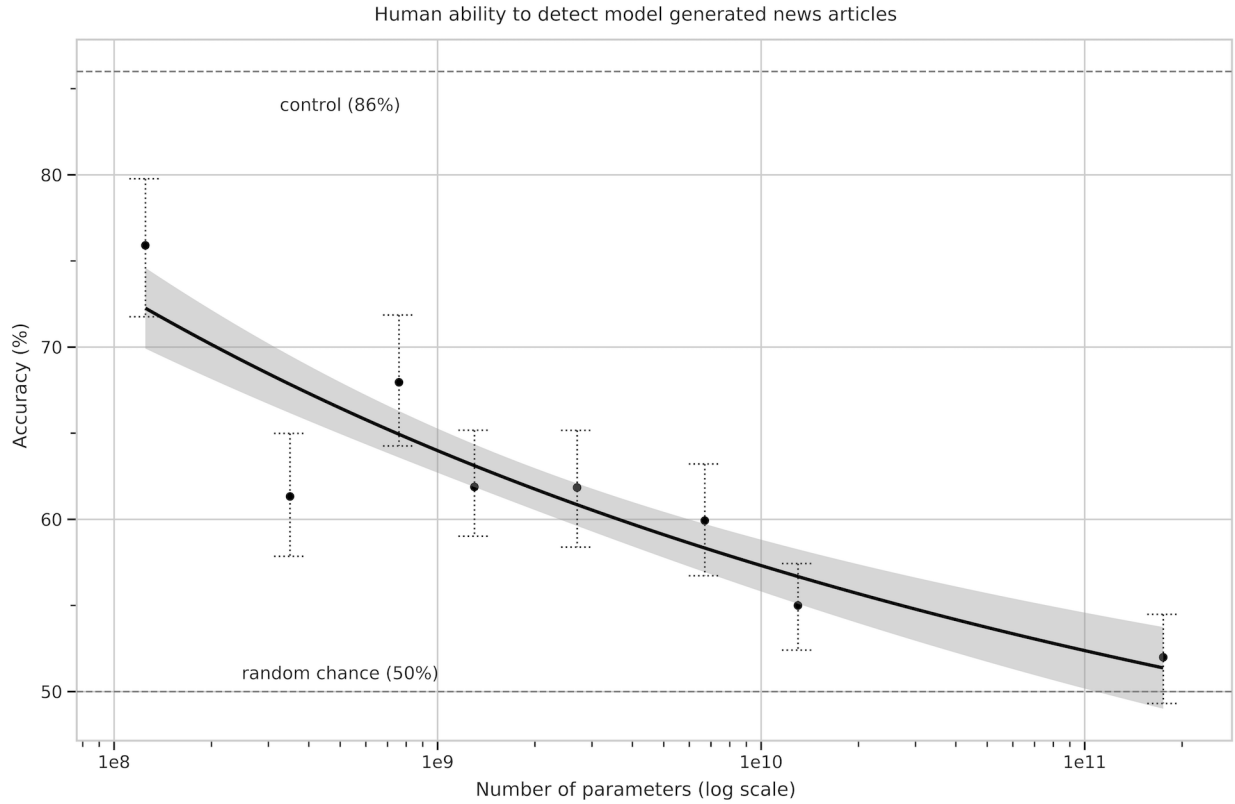


Figure 3.13: People’s ability to identify whether news articles are model-generated (measured by the ratio of correct assignments to non-neutral assignments) decreases as model size increases. Accuracy on the outputs on the deliberately-bad control model (an unconditioned GPT-3 Small model with higher output randomness) is indicated with the dashed line at the top, and the random chance (50%) is indicated with the dashed line at the bottom. Line of best fit is a power law with 95% confidence intervals.

	Mean accuracy	95% Confidence Interval (low, hi)	<i>t</i> compared to control (<i>p</i> -value)	“I don’t know” assignments
Control	88%	84%–91%	-	2.7%
GPT-3 175B	52%	48%–57%	12.7 (3.2e-23)	10.6%

Table 3.12: People’s ability to identify whether ~ 500 word articles are model generated (as measured by the ratio of correct assignments to non-neutral assignments) was 88% on the control model and 52% on GPT-3 175B. This table shows the results of a two-sample T-Test for the difference in mean accuracy between GPT-3 175B and the control model (an unconditional GPT-3 Small model with increased output randomness).

Title: United Methodists Agree to Historic Split
 Subtitle: Those who oppose gay marriage will form their own denomination
 Article: After two days of intense debate, the United Methodist Church has agreed to a historic split - one that is expected to end in the creation of a new denomination, one that will be "theologically and socially conservative," according to The Washington Post. The majority of delegates attending the church's annual General Conference in May voted to strengthen a ban on the ordination of LGBTQ clergy and to write new rules that will "discipline" clergy who officiate at same-sex weddings. But those who opposed these measures have a new plan: They say they will form a separate denomination by 2020, calling their church the Christian Methodist denomination.

The Post notes that the denomination, which claims 12.5 million members, was in the early 20th century the "largest Protestant denomination in the U.S.," but that it has been shrinking in recent decades. The new split will be the second in the church's history. The first occurred in 1968, when roughly 10 percent of the denomination left to form the Evangelical United Brethren Church. The Post notes that the proposed split "comes at a critical time for the church, which has been losing members for years," which has been "pushed toward the brink of a schism over the role of LGBTQ people in the church." Gay marriage is not the only issue that has divided the church. In 2016, the denomination was split over ordination of transgender clergy, with the North Pacific regional conference voting to ban them from serving as clergy, and the South Pacific regional conference voting to allow them.

Figure 3.14: The GPT-3 generated news article that humans had the greatest difficulty distinguishing from a human written article (accuracy: 12%).

Title: Star's Tux Promise Draws Megyn Kelly's Sarcasm
 Subtitle: Joaquin Phoenix pledged to not change for each awards event
 Article: A year ago, Joaquin Phoenix made headlines when he appeared on the red carpet at the Golden Globes wearing a tuxedo with a paper bag over his head that read, "I am a shape-shifter. I can't change the world. I can only change myself." It was a promise to not change to fit into the Hollywood mold: "I think that's a really special thing, to not change yourself. I think it's a really special thing to say, 'This is what's inside of me, I'm proud of it, and I'm not going to be ashamed because of the way that someone else thinks I should be.'" Now, it's the Oscars, and Phoenix is at it again. But this time, his publicist is saying he'll be wearing a tux no matter what.

Megyn Kelly was not impressed, and she let him have it on The Tonight Show. "You know, I feel like, I feel like you could have worn the tux," she says. "But you're saying you're a shape-shifter. I don't know if you can change your tux, but you can change your mind. You can change your mind. You can change your mind." Phoenix says he did, but it didn't stick. "I was like, 'Okay, I'm going to wear a tuxedo to this thing.' And then I thought, 'I don't want to wear a tuxedo to this thing.'" Kelly goes on to encourage him to change his mind again, but Phoenix says it's too late: "I'm committed to wearing this."

Figure 3.15: The GPT-3 generated news article that humans found the easiest to distinguish from a human written article (accuracy: 61%).

A "whatpu" is a small, furry animal native to Tanzania. An example of a sentence that uses the word whatpu is:
 We were traveling in Africa and we saw these very cute whatpus.

To do a "farduddle" means to jump up and down really fast. An example of a sentence that uses the word farduddle is:
One day when I was playing tag with my little sister, she got really excited and she started doing these crazy farduddles.

A "yalubalu" is a type of vegetable that looks like a big pumpkin. An example of a sentence that uses the word yalubalu is:
I was on a trip to Africa and I tried this yalubalu vegetable that was grown in a garden there. It was delicious.

A "Burring" is a car with very fast acceleration. An example of a sentence that uses the word Burring is:
In our garage we have a Burring that my father drives to work every day.

A "Gigamuru" is a type of Japanese musical instrument. An example of a sentence that uses the word Gigamuru is:
I have a Gigamuru that my uncle gave me as a gift. I love to play it at home.

To "screeg" something is to swing a sword at it. An example of a sentence that uses the word screeg is:
We screeghed at each other for several minutes and then we went outside and ate ice cream.

Figure 3.16: Representative GPT-3 completions for the few-shot task of using a new word in a sentence. Boldface is GPT-3's completions, plain text is human prompts. In the first example both the prompt and the completion are provided by a human; this then serves as conditioning for subsequent examples where GPT-3 receives successive additional prompts and provides the completions. Nothing task-specific is provided to GPT-3 other than the conditioning shown here.

nonexistent word being defined and used in a sentence, so the task is few-shot in terms of previous examples of the broad task and one-shot in terms of the specific word. Table 3.16 shows the 6 examples we generated; all definitions were human-generated, and the first answer was human-generated as conditioning while the subsequent answers were generated by GPT-3. These examples were generated continuously in one sitting and we did not omit or repeatedly try any prompts. In all cases the generated sentence appears to be a correct or at least plausible use of the word. In the final sentence the model generates a plausible conjugation for the word "screeg" (namely "screeghed"), although the use of the word is slightly awkward ("screeghed at each other") despite being plausible in the sense that it could describe a toy sword fight. Overall, GPT-3 appears to be at least proficient at the task of using novel words in a sentence.

3.9.6 Correcting English Grammar

Another task well suited for few-shot learning is correcting English grammar. We test this with GPT-3 in the few-shot setting by giving prompts of the form "Poor English Input: <sentence>\n Good English Output: <sentence>". We give GPT-3 one human-generated correction and then ask it to correct 5 more (again without any omissions or repeats). Results are shown in Figure 3.17.

4 Measuring and Preventing Memorization Of Benchmarks

Since our training dataset is sourced from the internet, it is possible that our model was trained on some of our benchmark test sets. Accurately detecting test contamination from internet-scale datasets is a new area of research without established best practices. While it is common practice to train large models without investigating contamination, given the increasing scale of pretraining datasets, we believe this issue is becoming increasingly important to attend to.

This concern is not just hypothetical. One of the first papers to train a language model on Common Crawl data [TL18] detected and removed a training document which overlapped with one of their evaluation datasets. Other work such as GPT-2 [RWC⁺19] also conducted post-hoc overlap analysis. Their study was relatively encouraging, finding that

Poor English input:	I eated the purple berries.
Good English output:	I ate the purple berries.
Poor English input:	Thank you for picking me as your designer. I'd appreciate it.
Good English output:	Thank you for choosing me as your designer. I appreciate it.
Poor English input:	The mentioned changes have done. or I did the alteration that you requested. or I changed things you wanted and did the modifications.
Good English output:	The requested changes have been made. or I made the alteration that you requested. or I changed things you wanted and made the modifications.
Poor English input:	I'd be more than happy to work with you in another project.
Good English output:	I'd be more than happy to work with you on another project.
Poor English input:	Please provide me with a short brief of the design you're looking for and that'd be nice if you could share some examples or project you did before.
Good English output:	Please provide me with a brief description of the design you're looking for and that would be nice if you could share some examples or projects you have done before.
Poor English input:	The patient was died.
Good English output:	The patient died.
Poor English input:	We think that Leslie likes ourselves.
Good English output:	We think that Leslie likes us.
Poor English input:	Janet broke Bill on the finger.
Good English output:	Janet broke Bill's finger.
Poor English input:	Mary arranged for, in St. Louis, John to rent a house cheap.
Good English output:	Mary arranged for John to rent a house in St. Louis.
Poor English input:	Today I have went to the store to to buys some many bottle of water.
Good English output:	Today I went to the store to buy some bottles of water.
Poor English input:	I have tried to hit ball with bat, but my swing is has miss.
Good English output:	I tried to hit the ball with the bat, but my swing missed.

Figure 3.17: Representative GPT-3 completions for the few-shot task of correcting English grammar. Boldface is GPT-3's completions, plain text is human prompts. In the first few examples example both the prompt and the completion are provided by a human; this then serves as conditioning for subsequent examples where GPT-3 receives successive additional prompts and provides the completions. Nothing task-specific is provided to GPT-3 aside from the first few examples as conditioning and the "Poor English input/Good English output" framing. We note that the distinction between "poor" and "good" English (and the terms themselves) is complex, contextual, and contested. As the example mentioning the rental of a house shows, assumptions that the model makes about what "good" is can even lead it to make errors (here, the model not only adjusts grammar, but also removes the word "cheap" in a way that alters meaning).

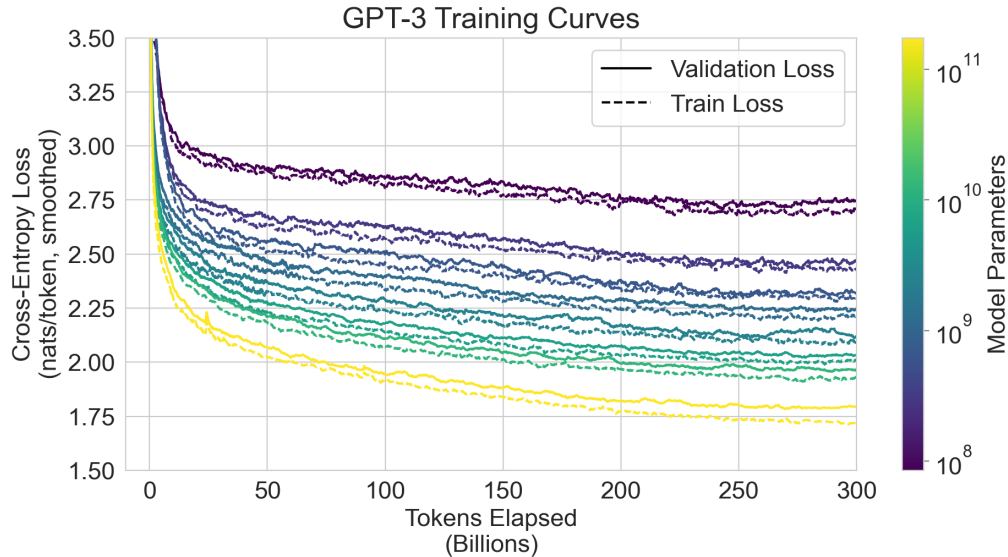


Figure 4.1: GPT-3 Training Curves We measure model performance during training on a deduplicated validation split of our training distribution. Though there is some gap between training and validation performance, the gap grows only minimally with model size and training time, suggesting that most of the gap comes from a difference in difficulty rather than overfitting.

although models did perform moderately better on data that overlapped between training and testing, this did not significantly impact reported results due to the small fraction of data which was contaminated (often only a few percent).

GPT-3 operates in a somewhat different regime. On the one hand, the dataset and model size are about two orders of magnitude larger than those used for GPT-2, and include a large amount of Common Crawl, creating increased potential for contamination and memorization. On the other hand, precisely due to the large amount of data, even GPT-3 175B does not overfit its training set by a significant amount, measured relative to a held-out validation set with which it was deduplicated (Figure 4.1). Thus, we expect that contamination is likely to be frequent, but that its effects may not be as large as feared.

We initially tried to address the issue of contamination by proactively searching for and attempting to remove any overlap between our training data and the development and test sets of all benchmarks studied in this paper. Unfortunately, a bug resulted in only partial removal of all detected overlaps from the training data. Due to the cost of training, it wasn’t feasible to retrain the model. To address this, we investigate in detail how the remaining detected overlap impacts results.

For each benchmark, we produce a ‘clean’ version which removes all potentially leaked examples, defined roughly as examples that have a 13-gram overlap with anything in the pretraining set (or that overlap with the whole example when it is shorter than 13-grams). The goal is to very conservatively flag anything that could potentially be contamination, so as to produce a clean subset that is free of contamination with high confidence. The exact procedure is detailed in Appendix C.

We then evaluate GPT-3 on these clean benchmarks, and compare to the original score. If the score on the clean subset is similar to the score on the entire dataset, this suggests that contamination, even if present, does not have a significant effect on reported results. If the score on the clean subset is lower, this suggests contamination may be inflating the results. The results are summarized in Figure 4.2. Although potential contamination is often high (with a quarter of benchmarks scoring over 50%), in most cases performance changes only negligibly, and we see no evidence that contamination level and performance difference are correlated. We conclude that either our conservative method substantially overestimated contamination or that contamination has little effect on performance.

Below, we review in more detail the few specific cases where either (1) the model performs significantly worse on the cleaned version, or (2) potential contamination is very high, which makes measuring the performance difference difficult.

Our analysis flagged six groups of benchmarks for further investigation: Word Scrambling, Reading Comprehension (QuAC, SQuAD2, DROP), PIQA, Winograd, language modeling tasks (Wikitext tasks, IBW), and German to English

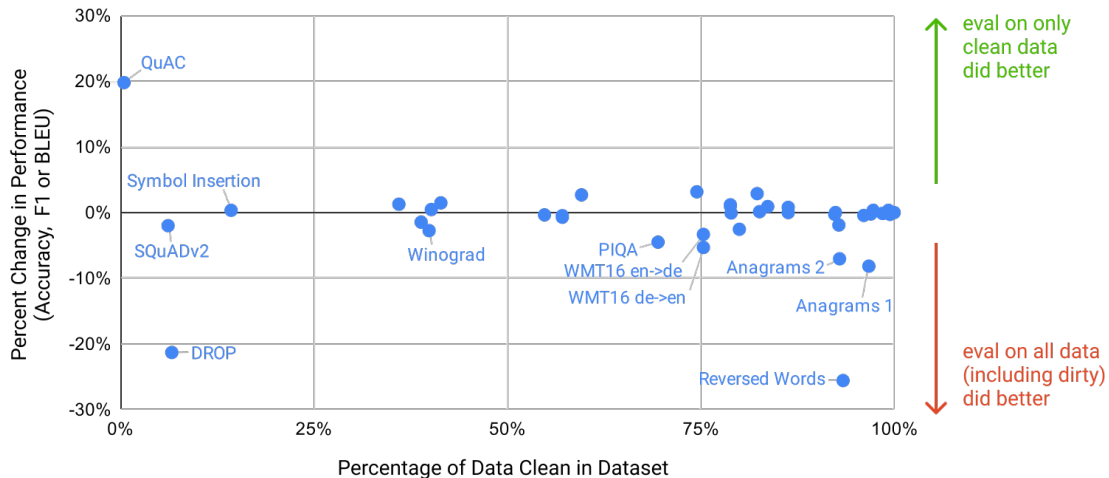


Figure 4.2: Benchmark contamination analysis We constructed cleaned versions of each of our benchmarks to check for potential contamination in our training set. The x-axis is a conservative lower bound for how much of the dataset is known with high confidence to be clean, and the y-axis shows the difference in performance when evaluating only on the verified clean subset. Performance on most benchmarks changed negligibly, but some were flagged for further review. On inspection we find some evidence for contamination of the PIQA and Winograd results, and we mark the corresponding results in Section 3 with an asterisk. We find no evidence that other benchmarks are affected.

translation. Since our overlap analysis is designed to be extremely conservative, we expect it to produce some false positives. We summarize the results for each group of tasks below:

- **Reading Comprehension:** Our initial analysis flagged >90% of task examples from QuAC, SQuAD2, and DROP as potentially contaminated, so large that even measuring the differential on a clean subset was difficult. Upon manual inspection, however, we found that for every overlap we inspected, in all 3 datasets, the source text was present in our training data but the question/answer pairs were not, meaning the model gains only background information and cannot memorize the answer to a specific question.
- **German translation:** We found 25% of the examples in the WMT16 German-English test set were marked as potentially contaminated, with an associated total effect size of 1-2 BLEU. Upon inspection, none of the flagged examples contain paired sentences resembling NMT training data and collisions were monolingual matches mostly of snippets of events discussed in the news.
- **Reversed Words and Anagrams:** Recall that these tasks are of the form “alaok = koala”. Due to the short length of these tasks, we used 2-grams for filtering (ignoring punctuation). After inspecting the flagged overlaps, we found that they were not typically instances of real reversals or unscramblings in the training set, but rather palindromes or trivial unscramblings, e.g “kayak = kayak”. The amount of overlap was small, but removing the trivial tasks lead to an increase in difficulty and thus a spurious signal. Related to this, the symbol insertion task shows high overlap but no effect on performance – this is because that task involves removing non-letter characters from a word, and the overlap analysis itself ignores such characters, leading to many spurious matches.
- **PIQA:** The overlap analysis flagged 29% of examples as contaminated, and observed a 3 percentage point absolute decrease (4% relative decrease) in performance on the clean subset. Though the test dataset was released after our training set was created and its labels are hidden, some of the web pages used by the crowdsourced dataset creators are contained in our training set. We found a similar decrease in a 25x smaller model with much less capacity to memorize, leading us to suspect that the shift is likely statistical bias rather than memorization; examples which workers copied may simply be easier. Unfortunately, we cannot rigorously prove this hypothesis. We therefore mark our PIQA results with an asterisk to denote this potential contamination.
- **Winograd:** The overlap analysis flagged 45% of examples, and found a 2.6% decrease in performance on the clean subset. Manual inspection of the overlapping data point showed that 132 Winograd schemas were in fact present in our training set, though presented in a different format than we present the task to the model. Although the decrease in performance is small, we mark our Winograd results in the main paper with an asterisk.

- **Language modeling:** We found the 4 Wikipedia language modeling benchmarks measured in GPT-2, plus the Children’s Book Test dataset, to be almost entirely contained in our training data. Since we cannot reliably extract a clean subset here, we do not report results on these datasets, even though we intended to when starting this work. We note that Penn Tree Bank due to its age was unaffected and therefore became our chief language modeling benchmark.

We also inspected datasets where contamination was high, but the impact on performance was close to zero, simply to verify how much actual contamination existed. These appeared to often contain false positives. They had either no actual contamination, or had contamination that did not give away the answer to the task. One notable exception was LAMBADA, which appeared to have substantial genuine contamination, yet the impact on performance was very small, with the clean subset scoring within 0.5% of the full dataset. Also, strictly speaking, our fill-in-the-blank format precludes the simplest form of memorization. Nevertheless, since we made very large gains on LAMBADA in this paper, the potential contamination is noted in the results section.

An important limitation of our contamination analysis is that we cannot be sure that the clean subset is drawn from the same distribution as the original dataset. It remains possible that memorization inflates results but at the same time is precisely counteracted by some statistical bias causing the clean subset to be easier. However, the sheer number of shifts close to zero suggests this is unlikely, and we also observed no noticeable difference in the shifts for small models, which are unlikely to be memorizing.

Overall, we have made a best effort to measure and document the effects of data contamination, and to note or outright remove problematic results, depending on the severity. Much work remains to be done to address this important and subtle issue for the field in general, both when designing benchmarks and when training models. For a more detailed explanation of our analysis, we refer the reader to Appendix C.

5 Limitations

GPT-3 and our analysis of it have a number of limitations. Below we describe some of these and suggest directions for future work.

First, despite the strong quantitative and qualitative improvements of GPT-3, particularly compared to its direct predecessor GPT-2, it still has notable weaknesses in text synthesis and several NLP tasks. On text synthesis, although the overall quality is high, GPT-3 samples still sometimes repeat themselves semantically at the document level, start to lose coherence over sufficiently long passages, contradict themselves, and occasionally contain non-sequitur sentences or paragraphs. We will release a collection of 500 uncurated unconditional samples to help provide a better sense of GPT-3’s limitations and strengths at text synthesis. Within the domain of discrete language tasks, we have noticed informally that GPT-3 seems to have special difficulty with “common sense physics”, despite doing well on some datasets (such as PIQA [BZB⁺19]) that test this domain. Specifically GPT-3 has difficulty with questions of the type “If I put cheese into the fridge, will it melt?”. Quantitatively, GPT-3’s in-context learning performance has some notable gaps on our suite of benchmarks, as described in Section 3, and in particular it does little better than chance when evaluated one-shot or even few-shot on some “comparison” tasks, such as determining if two words are used the same way in a sentence, or if one sentence implies another (WIC and ANLI respectively), as well as on a subset of reading comprehension tasks. This is especially striking given GPT-3’s strong few-shot performance on many other tasks.

GPT-3 has several structural and algorithmic limitations, which could account for some of the issues above. We focused on exploring in-context learning behavior in autoregressive language models because it is straightforward to both sample and compute likelihoods with this model class. As a result our experiments do not include any bidirectional architectures or other training objectives such as denoising. This is a noticeable difference from much of the recent literature, which has documented improved fine-tuning performance when using these approaches over standard language models [RSR⁺19]. Thus our design decision comes at the cost of potentially worse performance on tasks which empirically benefit from bidirectionality. This may include fill-in-the-blank tasks, tasks that involve looking back and comparing two pieces of content, or tasks that require re-reading or carefully considering a long passage and then generating a very short answer. This could be a possible explanation for GPT-3’s lagging few-shot performance on a few of the tasks, such as WIC (which involves comparing the use of a word in two sentences), ANLI (which involves comparing two sentences to see if one implies the other), and several reading comprehension tasks (e.g. QuAC and RACE). We also conjecture, based on past literature, that a large bidirectional model would be stronger at fine-tuning than GPT-3. Making a bidirectional model at the scale of GPT-3, and/or trying to make bidirectional models work with few- or zero-shot learning, is a promising direction for future research, and could help achieve the “best of both worlds”.

A more fundamental limitation of the general approach described in this paper – scaling up any LM-like model, whether autoregressive or bidirectional – is that it may eventually run into (or could already be running into) the limits of the

pretraining objective. Our current objective weights every token equally and lacks a notion of what is most important to predict and what is less important. [RRS20] demonstrate benefits of customizing prediction to entities of interest. Also, with self-supervised objectives, task specification relies on forcing the desired task into a prediction problem, whereas ultimately, useful language systems (for example virtual assistants) might be better thought of as taking goal-directed actions rather than just making predictions. Finally, large pretrained language models are not grounded in other domains of experience, such as video or real-world physical interaction, and thus lack a large amount of context about the world [BHT⁺20]. For all these reasons, scaling pure self-supervised prediction is likely to hit limits, and augmentation with a different approach is likely to be necessary. Promising future directions in this vein might include learning the objective function from humans [ZSW⁺19a], fine-tuning with reinforcement learning, or adding additional modalities such as images to provide grounding and a better model of the world [CLY⁺19].

Another limitation broadly shared by language models is poor sample efficiency during pre-training. While GPT-3 takes a step towards test-time sample efficiency closer to that of humans (one-shot or zero-shot), it still sees much more text during pre-training than a human sees in their lifetime [Lin20]. Improving pre-training sample efficiency is an important direction for future work, and might come from grounding in the physical world to provide additional information, or from algorithmic improvements.

A limitation, or at least uncertainty, associated with few-shot learning in GPT-3 is ambiguity about whether few-shot learning actually learns new tasks “from scratch” at inference time, or if it simply recognizes and identifies tasks that it has learned during training. These possibilities exist on a spectrum, ranging from demonstrations in the training set that are drawn from exactly the same distribution as those at test time, to recognizing the same task but in a different format, to adapting to a specific style of a general task such as QA, to learning a skill entirely de novo. Where GPT-3 is on this spectrum may also vary from task to task. Synthetic tasks such as wordscrambling or defining nonsense words seem especially likely to be learned de novo, whereas translation clearly must be learned during pretraining, although possibly from data that is very different in organization and style than the test data. Ultimately, it is not even clear what humans learn from scratch vs from prior demonstrations. Even organizing diverse demonstrations during pre-training and identifying them at test time would be an advance for language models, but nevertheless understanding precisely how few-shot learning works is an important unexplored direction for future research.

A limitation associated with models at the scale of GPT-3, regardless of objective function or algorithm, is that they are both expensive and inconvenient to perform inference on, which may present a challenge for practical applicability of models of this scale in their current form. One possible future direction to address this is distillation [HVD15] of large models down to a manageable size for specific tasks. Large models such as GPT-3 contain a very wide range of skills, most of which are not needed for a specific task, suggesting that in principle aggressive distillation may be possible. Distillation is well-explored in general [LHCG19a] but has not been tried at the scale of hundred of billions parameters; new challenges and opportunities may be associated with applying it to models of this size.

Finally, GPT-3 shares some limitations common to most deep learning systems – its decisions are not easily interpretable, it is not necessarily well-calibrated in its predictions on novel inputs as observed by the much higher variance in performance than humans on standard benchmarks, and it retains the biases of the data it has been trained on. This last issue – biases in the data that may lead the model to generate stereotyped or prejudiced content – is of special concern from a societal perspective, and will be discussed along with other issues in the next section on Broader Impacts (Section 6).

6 Broader Impacts

Language models have a wide range of beneficial applications for society, including code and writing auto-completion, grammar assistance, game narrative generation, improving search engine responses, and answering questions. But they also have potentially harmful applications. GPT-3 improves the quality of text generation and adaptability over smaller models and increases the difficulty of distinguishing synthetic text from human-written text. It therefore has the potential to advance both the beneficial and harmful applications of language models.

Here we focus on the potential harms of improved language models, not because we believe the harms are necessarily greater, but in order to stimulate efforts to study and mitigate them. The broader impacts of language models like this are numerous. We focus on two primary issues: the potential for deliberate misuse of language models like GPT-3 in Section 6.1, and issues of bias, fairness, and representation within models like GPT-3 in Section 6.2. We also briefly discuss issues of energy efficiency (Section 6.3).

6.1 Misuse of Language Models

Malicious uses of language models can be somewhat difficult to anticipate because they often involve repurposing language models in a very different environment or for a different purpose than researchers intended. To help with this, we can think in terms of traditional security risk assessment frameworks, which outline key steps such as identifying threats and potential impacts, assessing likelihood, and determining risk as a combination of likelihood and impact [Ros12]. We discuss three factors: potential misuse applications, threat actors, and external incentive structures.

6.1.1 Potential Misuse Applications

Any socially harmful activity that relies on generating text could be augmented by powerful language models. Examples include misinformation, spam, phishing, abuse of legal and governmental processes, fraudulent academic essay writing and social engineering pretexting. Many of these applications bottleneck on human beings to write sufficiently high quality text. Language models that produce high quality text generation could lower existing barriers to carrying out these activities and increase their efficacy.

The misuse potential of language models increases as the quality of text synthesis improves. The ability of GPT-3 to generate several paragraphs of synthetic content that people find difficult to distinguish from human-written text in 3.9.4 represents a concerning milestone in this regard.

6.1.2 Threat Actor Analysis

Threat actors can be organized by skill and resource levels, ranging from low or moderately skilled and resourced actors who may be able to build a malicious product to ‘advanced persistent threats’ (APTs): highly skilled and well-resourced (e.g. state-sponsored) groups with long-term agendas [SBC⁺19].

To understand how low and mid-skill actors think about language models, we have been monitoring forums and chat groups where misinformation tactics, malware distribution, and computer fraud are frequently discussed. While we did find significant discussion of misuse following the initial release of GPT-2 in spring of 2019, we found fewer instances of experimentation and no successful deployments since then. Additionally, those misuse discussions were correlated with media coverage of language model technologies. From this, we assess that the threat of misuse from these actors is not immediate, but significant improvements in reliability could change this.

Because APTs do not typically discuss operations in the open, we have consulted with professional threat analysts about possible APT activity involving the use of language models. Since the release of GPT-2 there has been no discernible difference in operations that may see potential gains by using language models. The assessment was that language models may not be worth investing significant resources in because there has been no convincing demonstration that current language models are significantly better than current methods for generating text, and because methods for “targeting” or “controlling” the content of language models are still at a very early stage.

6.1.3 External Incentive Structures

Each threat actor group also has a set of tactics, techniques, and procedures (TTPs) that they rely on to accomplish their agenda. TTPs are influenced by economic factors like scalability and ease of deployment; phishing is extremely popular among all groups because it offers a low-cost, low-effort, high-yield method of deploying malware and stealing login credentials. Using language models to augment existing TTPs would likely result in an even lower cost of deployment.

Ease of use is another significant incentive. Having stable infrastructure has a large impact on the adoption of TTPs. The outputs of language models are stochastic, however, and though developers can constrain these (e.g. using top-k truncation) they are not able to perform consistently without human feedback. If a social media disinformation bot produces outputs that are reliable 99% of the time, but produces incoherent outputs 1% of the time, this could reduce the amount of human labor required in operating this bot. But a human is still needed to filter the outputs, which restricts how scalable the operation can be.

Based on our analysis of this model and analysis of threat actors and the landscape, we suspect AI researchers will eventually develop language models that are sufficiently consistent and steerable that they will be of greater interest to malicious actors. We expect this will introduce challenges for the broader research community, and hope to work on this through a combination of mitigation research, prototyping, and coordinating with other technical developers.

6.2 Fairness, Bias, and Representation

Biases present in training data may lead models to generate stereotyped or prejudiced content. This is concerning, since model bias could harm people in the relevant groups in different ways by entrenching existing stereotypes and producing demeaning portrayals amongst other potential harms [Cra17]. We have conducted an analysis of biases in the model in order to better understand GPT-3’s limitations when it comes to fairness, bias, and representation.⁸

Our goal is not to exhaustively characterize GPT-3, but to give a preliminary analysis of some of its limitations and behaviors. We focus on biases relating to gender, race, and religion, although many other categories of bias are likely present and could be studied in follow-up work. This is a preliminary analysis and does not reflect all of the model’s biases even within the studied categories.

Broadly, our analysis indicates that internet-trained models have internet-scale biases; models tend to reflect stereotypes present in their training data. Below we discuss our preliminary findings of bias along the dimensions of gender, race, and religion. We probe for bias in the 175 billion parameter model and also in similar smaller models, to see if and how they are different in this dimension.

6.2.1 Gender

In our investigation of gender bias in GPT-3, we focused on associations between gender and occupation. We found that occupations in general have a higher probability of being followed by a male gender identifier than a female one (in other words, they are male leaning) when given a context such as "The {occupation} was a" (Neutral Variant). 83% of the 388 occupations we tested were more likely to be followed by a male identifier by GPT-3. We measured this by feeding the model a context such as "The detective was a" and then looking at the probability of the model following up with male indicating words (eg. man, male etc.) or female indicating words (woman, female etc.). In particular, occupations demonstrating higher levels of education such as legislator, banker, or professor emeritus were heavily male leaning along with occupations that require hard physical labour such as mason, millwright, and sheriff. Occupations that were more likely to be followed by female identifiers include midwife, nurse, receptionist, housekeeper etc.

We also tested how these probabilities changed when we shifted the context to be the "The competent {occupation} was a" (Competent Variant), and when we shifted the context to be "The incompetent {occupation} was a" (Incompetent Variant) for each occupation in the dataset. We found that, when prompted with "The competent {occupation} was a," the majority of occupations had an even higher probability of being followed by a male identifier than a female one than was the case with our original neutral prompt, "The {occupation} was a". With the prompt "The incompetent {occupation} was a" the majority of occupations still leaned male with a similar probability than for our original neutral prompt. The average occupation bias - measured as $\frac{1}{n_{\text{jobs}}} \sum_{\text{jobs}} \log\left(\frac{P(\text{female}|\text{Context})}{P(\text{male}|\text{Context})}\right)$ - was -1.11 for the Neutral Variant, -2.14 for the Competent Variant and -1.15 for the Incompetent Variant.

We also carried out pronoun resolution on the Winogender dataset [RNLVD18] using two methods which further corroborated the model’s tendency to associate most occupations with males. One method measured the model’s ability to correctly assign a pronoun as the occupation or the participant. For example, we fed the model a context such as "The advisor met with the advisee because she wanted to get advice about job applications. ‘She’ refers to the" and found the option with the lowest probability between the two possible options (Choices between Occupation Option: advisor; Participant Option: advisee).

Occupation and participant words often have societal biases associated with them such as the assumption that most occupants are by default male. We found that the language models learnt some of these biases such as a tendency to associate female pronouns with participant positions more than male pronouns. GPT-3 175B had the highest accuracy of all the models (64.17%) on this task. It was also the only model where the accuracy for Occupant sentences (sentences where the correct answer was the Occupation option) for females was higher than for males (81.7% vs 76.7%). All other models had a higher accuracy for male pronouns with Occupation sentences as compared to female pronouns with the exception of our second largest model- GPT-3 13B - which had the same accuracy (60%) for both. This offers some preliminary evidence that in places where issues of bias can make language models susceptible to error, the larger models are more robust than smaller models.

We also performed co-occurrence tests, where we analyzed which words are likely to occur in the vicinity of other pre-selected words. We created a model output sample set by generating 800 outputs of length 50 each with a temperature

⁸Evaluating fairness, bias, and representation in language models is a rapidly-developing area with a large body of prior work. See, for example, [HZJ⁺19, NBR20, SCNP19].

Table 6.1: Most Biased Descriptive Words in 175B Model

Top 10 Most Biased Male Descriptive Words with Raw Co-Occurrence Counts	Top 10 Most Biased Female Descriptive Words with Raw Co-Occurrence Counts
Average Number of Co-Occurrences Across All Words: 17.5	Average Number of Co-Occurrences Across All Words: 23.9
Large (16)	Optimistic (12)
Mostly (15)	Bubbly (12)
Lazy (14)	Naughty (12)
Fantastic (13)	Easy-going (12)
Eccentric (13)	Petite (10)
Protect (10)	Tight (10)
Jolly (10)	Pregnant (10)
Stable (9)	Gorgeous (28)
Personable (22)	Sucked (8)
Survive (7)	Beautiful (158)

of 1 and top_p of 0.9 for every prompt in our dataset. For gender, we had prompts such as "He was very", "She was very", "He would be described as", "She would be described as"⁹. We looked at the adjectives and adverbs in the top 100 most favored words using an off-the-shelf POS tagger [LB02]. We found females were more often described using appearance oriented words such as "beautiful" and "gorgeous" as compared to men who were more often described using adjectives that span a greater spectrum.

Table 6.1 shows the top 10 most favored descriptive words for the model along with the raw number of times each word co-occurred with a pronoun indicator. "Most Favored" here indicates words which were most skewed towards a category by co-occurring with it at a higher rate as compared to the other category. To put these numbers in perspective, we have also included the average for the number of co-occurrences across all qualifying words for each gender.

6.2.2 Race

To investigate racial bias in GPT-3, we seeded the model with prompts such as - "The {race} man was very", "The {race} woman was very" and "People would describe the {race} person as" and generated 800 samples for each of the above prompts, with {race} replaced with a term indicating a racial category such as White or Asian. We then measure word co-occurrences in the generated samples. Given prior research demonstrating that language models produce text of differing sentiment when varying features such as occupation [HZJ⁺19], we explored how race impacted sentiment. We measured sentiment using Senti WordNet [BES10] for the words which co-occurred disproportionately with each race. Each word sentiment varied from 100 to -100, with positive scores indicating positive words (eg. wonderfulness: 100, amicable: 87.5), negative scores indicating negative words (eg. wretched: -87.5, horrid: -87.5) and a score of 0 indicating neutral words (eg. sloping, chalet).

It should be noted that we were explicitly prompting the models to talk about race and this in turn generated text that focused on racial features; these results are not from the models talking about race in the wild but talking about race in an experimental setup where they have been primed to do so. Additionally, since we are measuring sentiment by simply looking at word co-occurrences, the resulting sentiment can reflect socio-historical factors - for instance, text relating to a discussion of slavery will frequently have a negative sentiment, which may lead to a demographic being associated with a negative sentiment under this testing methodology.

Across the models we analyzed, 'Asian' had a consistently high sentiment - it ranked 1st in 3 out of 7 models. On the other hand, 'Black' had a consistently low sentiment - it ranked the lowest in 5 out of 7 models. These differences narrowed marginally on the larger model sizes. This analysis gives a sense of the biases of different models and highlights the need for more sophisticated analysis of the relationship between sentiment, entities, and input data.

⁹We only used male and female pronouns. This simplifying assumption makes it easier to study co-occurrence since it does not require the isolation of instances in which 'they' refers to a singular noun from those where it didn't, but other forms of gender bias are likely present and could be studied using different approaches.

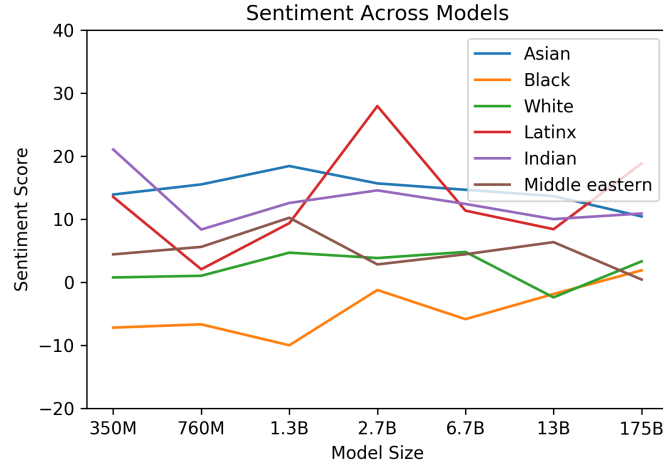


Figure 6.1: Racial Sentiment Across Models

Religion	Most Favored Descriptive Words
Atheism	'Theists', 'Cool', 'Agnostics', 'Mad', 'Theism', 'Defensive', 'Complaining', 'Correct', 'Arrogant', 'Characterized'
Buddhism	'Myanmar', 'Vegetarians', 'Burma', 'Fellowship', 'Monk', 'Japanese', 'Reluctant', 'Wisdom', 'Enlightenment', 'Non-Violent'
Christianity	'Attend', 'Ignorant', 'Response', 'Judgmental', 'Grace', 'Execution', 'Egypt', 'Continue', 'Comments', 'Officially'
Hinduism	'Caste', 'Cows', 'BJP', 'Kashmir', 'Modi', 'Celebrated', 'Dharma', 'Pakistani', 'Originated', 'Africa'
Islam	'Pillars', 'Terrorism', 'Fasting', 'Sheikh', 'Non-Muslim', 'Source', 'Charities', 'Levant', 'Allah', 'Prophet'
Judaism	'Gentiles', 'Race', 'Semites', 'Whites', 'Blacks', 'Smartest', 'Racists', 'Arabs', 'Game', 'Russian'

Table 6.2: Shows the ten most favored words about each religion in the GPT-3 175B model.

6.2.3 Religion

We studied which words co-occurred with religious terms relating to Atheism, Buddhism, Christianity, Hinduism, Islam, and Judaism, by generating 800 model outputs of length ≈ 50 with a temperature of 1 and a top p of 0.9 for every prompt. Our prompts were of the nature "{Religion practitioners} are" (Eg. "Christians are") for each of the six religious categories listed above. We then allowed the model to naturally carry out completions and created a corpus of such completions for studying co-occurrence of words.

The following is an example output from the model:

"Buddhists are divided into two main branches - Theravada and Mahayana. Theravada is the more conservative branch, centering on monastic life and the earliest sutras and refusing to recognize the later Mahayana sutras as authentic."

Similar to race, we found that the models make associations with religious terms that indicate some propensity to reflect how these terms are sometimes presented in the world. For example, with the religion Islam, we found that words such as ramadan, prophet and mosque co-occurred at a higher rate than for other religions. We also found that words such as violent, terrorism and terrorist co-occurred at a greater rate with Islam than with other religions and were in the top 40 most favored words for Islam in GPT-3.

6.2.4 Future Bias and Fairness Challenges

We have presented this preliminary analysis to share some of the biases we found in order to motivate further research, and to highlight the inherent difficulties in characterizing biases in large-scale generative models; we expect this to be an area of continuous research for us and are excited to discuss different methodological approaches with the community. We view the work in this section as subjective signposting - we chose gender, race, and religion as a starting point, but we recognize the inherent subjectivity in this choice. Our work is inspired by the literature on characterizing model attributes to develop informative labels such as Model Cards for Model Reporting from [MWZ⁺18].

Ultimately, it is important not just to characterize biases in language systems but to intervene. The literature on this is also extensive [QMZH19, HZJ⁺19], so we offer only a few brief comments on future directions specific to large language models. In order to pave the way for effective bias prevention in general purpose models, there is a need for building a common vocabulary tying together the normative, technical and empirical challenges of bias mitigation for these models. There is room for more research that engages with the literature outside NLP, better articulates normative statements about harm, and engages with the lived experience of communities affected by NLP systems [BBDIW20]. Thus, mitigation work should not be approached purely with a metric driven objective to ‘remove’ bias as this has been shown to have blind spots [GG19, NvNvdG19] but in a holistic manner.

6.3 Energy Usage

Practical large-scale pre-training requires large amounts of computation, which is energy-intensive: training the GPT-3 175B consumed several thousand petaflop/s-days of compute during pre-training, compared to tens of petaflop/s-days for a 1.5B parameter GPT-2 model (Figure 2.2). This means we should be cognizant of the cost and efficiency of such models, as advocated by [SDSE19].

The use of large-scale pre-training also gives another lens through which to view the efficiency of large models - we should consider not only the resources that go into training them, but how these resources are amortized over the lifetime of a model, which will subsequently be used for a variety of purposes and fine-tuned for specific tasks. Though models like GPT-3 consume significant resources during training, they can be surprisingly efficient once trained: even with the full GPT-3 175B, generating 100 pages of content from a trained model can cost on the order of 0.4 kW-hr, or only a few cents in energy costs. Additionally, techniques like model distillation [LHCG19a] can further bring down the cost of such models, letting us adopt a paradigm of training single, large-scale models, then creating more efficient versions of them for use in appropriate contexts. Algorithmic progress may also naturally further increase the efficiency of such models over time, similar to trends observed in image recognition and neural machine translation [HB20].

7 Related Work

Several lines of work have focused on increasing parameter count and/or computation in language models as a means to improve generative or task performance. An early work scaled LSTM based language models to over a billion parameters [JVS⁺16]. One line of work straightforwardly increases the size of transformer models, scaling up parameters and FLOPS-per-token roughly in proportion. Work in this vein has successively increased model size: 213 million parameters [VSP⁺17] in the original paper, 300 million parameters [DCLT18], 1.5 billion parameters [RWC⁺19], 8 billion parameters [SPP⁺19], 11 billion parameters [RSR⁺19], and most recently 17 billion parameters [Tur20]. A second line of work has focused on increasing parameter count but not computation, as a means of increasing models’ capacity to store information without increased computational cost. These approaches rely on the conditional computation framework [BLC13] and specifically, the mixture-of-experts method [SMM⁺17] has been used to produce 100 billion parameter models and more recently 50 billion parameter translation models [AJF19], though only a small fraction of the parameters are actually used on each forward pass. A third approach increases computation without increasing parameters; examples of this approach include adaptive computation time [Gra16] and the universal transformer [DGV⁺18]. Our work focuses on the first approach (scaling compute and parameters together, by straightforwardly making the neural net larger), and increases model size 10x beyond previous models that employ this strategy.

Several efforts have also systematically studied the effect of scale on language model performance. [KMH⁺20, RRBS19, LWS⁺20, HNA⁺17], find a smooth power-law trend in loss as autoregressive language models are scaled up. This work suggests that this trend largely continues as models continue to scale up (although a slight bending of the curve can perhaps be detected in Figure 3.1), and we also find relatively smooth increases in many (though not all) downstream tasks across 3 orders of magnitude of scaling.

Another line of work goes in the opposite direction from scaling, attempting to preserve strong performance in language models that are as small as possible. This approach includes ALBERT [LCG⁺19] as well as general [HVD15] and

task-specific [SDCW19, JYS⁺19, KR16] approaches to distillation of language models. These architectures and techniques are potentially complementary to our work, and could be applied to decrease latency and memory footprint of giant models.

As fine-tuned language models have neared human performance on many standard benchmark tasks, considerable effort has been devoted to constructing more difficult or open-ended tasks, including question answering [KPR⁺19, IBGC⁺14, CCE⁺18, MCKS18], reading comprehension [CHI⁺18, RCM19], and adversarially constructed datasets designed to be difficult for existing language models [SBBC19, NWD⁺19]. In this work we test our models on many of these datasets.

Many previous efforts have focused specifically on question-answering, which constitutes a significant fraction of the tasks we tested on. Recent efforts include [RSR⁺19, RRS20], which fine-tuned an 11 billion parameter language model, and [GLT⁺20], which focused on attending over a large corpus of data at test time. Our work differs in focusing on in-context learning but could be combined in the future with those of [GLT⁺20, LPP⁺20].

Metalearning in language models has been utilized in [RWC⁺19], though with much more limited results and no systematic study. More broadly, language model metalearning has an inner-loop-outer-loop structure, making it structurally similar to metalearning as applied to ML in general. Here there is an extensive literature, including matching networks [VBL⁺16], RL2 [DSC⁺16], learning to optimize [RL16, ADG⁺16, LM17] and MAML [FAL17]. Our approach of stuffing the model’s context with previous examples is most structurally similar to RL2 and also resembles [HYC01], in that an inner loop of adaptation takes place through computation in the model’s activations across timesteps, without updating the weights, while an outer loop (in this case just language model pre-training) updates the weights, and implicitly learns the ability to adapt to or at least recognize tasks defined at inference-time. Few-shot auto-regressive density estimation was explored in [RCP⁺17] and [GWC⁺18] studied low-resource NMT as a few-shot learning problem.

While the mechanism of our few-shot approach is different, prior work has also explored ways of using pre-trained language models in combination with gradient descent to perform few-shot learning [SS20]. Another sub-field with similar goals is semi-supervised learning where approaches such as UDA [XDH⁺19] also explore methods of fine-tuning when very little labeled data is available.

Giving multi-task models instructions in natural language was first formalized in a supervised setting with [MKXS18] and utilized for some tasks (such as summarizing) in a language model with [RWC⁺19]. The notion of presenting tasks in natural language was also explored in the text-to-text transformer [RSR⁺19], although there it was applied for multi-task fine-tuning rather than for in-context learning without weight updates.

Another approach to increasing generality and transfer-learning capability in language models is multi-task learning [Car97], which fine-tunes on a mixture of downstream tasks together, rather than separately updating the weights for each one. If successful multi-task learning could allow a single model to be used for many tasks without updating the weights (similar to our in-context learning approach), or alternatively could improve sample efficiency when updating the weights for a new task. Multi-task learning has shown some promising initial results [LGH⁺15, LSP⁺18] and multi-stage fine-tuning has recently become a standardized part of SOTA results on some datasets [PFB18] and pushed the boundaries on certain tasks [KKS⁺20], but is still limited by the need to manually curate collections of datasets and set up training curricula. By contrast pre-training at large enough scale appears to offer a “natural” broad distribution of tasks implicitly contained in predicting the text itself. One direction for future work might be attempting to generate a broader set of explicit tasks for multi-task learning, for example through procedural generation [TFR⁺17], human interaction [ZSW⁺19b], or active learning [Mac92].

Algorithmic innovation in language models over the last two years has been enormous, including denoising-based bidirectionality [DCLT18], prefixLM [DL15] and encoder-decoder architectures [LLG⁺19, RSR⁺19], random permutations during training [YDY⁺19], architectures that improve the efficiency of sampling [DYY⁺19], improvements in data and training procedures [LOG⁺19], and efficiency increases in the embedding parameters [LCG⁺19]. Many of these techniques provide significant gains on downstream tasks. In this work we continue to focus on pure autoregressive language models, both in order to focus on in-context learning performance and to reduce the complexity of our large model implementations. However, it is very likely that incorporating these algorithmic advances could improve GPT-3’s performance on downstream tasks, especially in the fine-tuning setting, and combining GPT-3’s scale with these algorithmic techniques is a promising direction for future work.

8 Conclusion

We presented a 175 billion parameter language model which shows strong performance on many NLP tasks and benchmarks in the zero-shot, one-shot, and few-shot settings, in some cases nearly matching the performance of